# [Demo] Muon Tomography: From Counts to Density

BANDAL Demo Workspace | Sample research note | 2026

Status: authored sample, not a real publication. All particle counts, viewing bins and transmission ratios below are synthetic. No detector was operated.

## Abstract

This fictional example uses five viewing bins to illustrate a transmission-count comparison. Every reference bin contains 1,000 invented events; corresponding target counts range from 650 to 930. The central bin has the lowest synthetic transmission ratio. The note demonstrates how a reader can connect numerical evidence to an interpretation without presenting invented measurements as scientific results.

## 1. Counting model

For each viewing bin, T = N_target / N_reference. Reference and target counts are assumed to have equal exposure for this demonstration. Lower transmission motivates a density question, but this sample does not solve an inverse problem or infer a physical density.

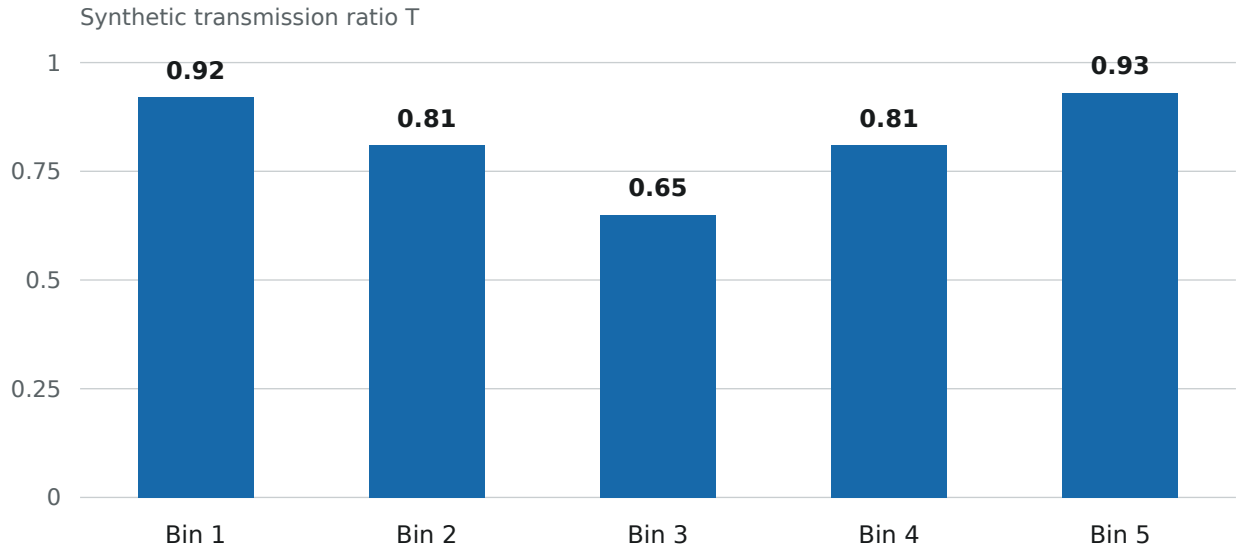



Figure 1. Transmission ratios from invented counts. Viewing bins are labels; no angle or physical distance is assigned.

# 2. Synthetic count table

| Viewing bin | Reference count | Target count | Transmission T |
|---|---|---|---|
| 1 | 1,000 | 920 | 0.92 |
| 2 | 1,000 | 810 | 0.81 |
| 3 | 1,000 | 650 | 0.65 |
| 4 | 1,000 | 810 | 0.81 |
| 5 | 1,000 | 930 | 0.93 |

Worked example for Bin 3: T = 650 / 1,000 = 0.65. The deficit relative to the invented reference is 350 events, or 35%. Those values are arithmetic outputs of synthetic inputs, not an observed anomaly.

## 3. Interpretation and limits

The central-bin deficit is a useful annotation target. Equal exposure is an explicit simplifying assumption. Detector acceptance, direction-dependent flux, background and counting uncertainty would all need treatment in a real experiment. Different geometric models could explain similar transmission patterns.

## 4. Suggested reading notes

Question: How would unequal exposure change the transmission calculation?
Claim to annotate: a count deficit alone does not determine density.
Connection: link this sample to the seismic uncertainty note to compare two indirect measurements.

## Provenance

Created specifically as a sample PDF for the BANDAL-hosted BluePrint demonstration. The dataset and document are authored demonstration material. No journal article, detector result or external source is represented.

Reader exercise: highlight the Bin 3 calculation and record which assumptions would be required before interpreting a real target.